

Bank entry and endogenous pricing power in the deposit market

Sonny Biswas* Kostas Koufopoulos[†]

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Abstract

Agents exert costly effort to become bankers and banks can pledge only a fraction of their output to depositors. Although banks compete in prices, limited pledgeability restricts leverage, which confers pricing power to banks, thereby enabling them to recover sunk costs associated with their initial effort. In contrast to conventional wisdom, but consistent with empirical evidence, equilibrium pricing power may be negatively correlated with bank concentration. While in partial equilibrium, stronger investment opportunities lead to higher bank leverage, accounting for banks' entry decision reverses this prediction. Despite a strictly positive entry cost, the laissez-faire features excessive entry, inviting regulatory intervention.

Keywords: Bank concentration, Competition, Excessive entry, Ex-post moral hazard

Jel codes: D40, G21, G28

*University of Bristol Business School, United Kingdom. s.biswas@bristol.ac.uk.

[†]University of Sussex Business School, United Kingdom. k.koufopoulos@sussex.ac.uk.

1 Introduction

Deposits serve as the primary funding source for banks, with the US deposit market exceeding \$17.5 trillion in 2024, accounting for over 80% of bank liabilities. Evidence consistently shows that banks possess substantial pricing power in deposit markets (see Forssbaeck and Shehzad 2015 and Beck et al. 2013 for global evidence and Drechsler et al. 2017 and Corbae and D’Erasmus 2021 for US data). While the four largest banks make up roughly two-fifths of the US deposit market, numerous regional banks and smaller institutions contribute to the competitive landscape. However, cross-sectional differentiation remains minimal in this market, as banks largely compete along a single dimension, prices (deposit rates). This raises a fundamental question: how do banks maintain pricing power in what appears to be a relatively homogeneous market?

We present a model of bank entry and price competition in the deposit market which features two informational frictions: an effort moral hazard problem where agents must exert costly effort, and a diversion constraint which limits banks’ ability to pledge future output to secure funding in the deposit market. Despite price competition among banks in their attempt to attract deposits, the limited pledgeability constraint prevents the standard Bertrand outcome from arising for all parameters other than a knife-edge case (effort is costless). As a result, the degree of competition in the deposit market is endogenous in our model. We derive the conditions under which the deposit market features monopolistic prices, oligopolistic prices, or perfectly competitive prices in equilibrium. Our analysis shows that the use of concentration-based measures of bank competition may result in misleading inferences about market power since high concentration can coexist with low pricing power. Although the cost of entry is strictly positive, bank entry may be excessive, and the regulator can eliminate this inefficiency by introducing an upper bound on the number of bank licences to be allocated.

Preview of the model and results.

In the model, agents who have an initial endowment may choose to exert costly effort to become bankers. Agents are identical to begin with and they become bankers with probability 1 if they exert effort. If they do not become bankers, agents delegate the man-

agement of their funds to bankers or put them in the storage technology. Banks invest their own endowment and the funds collected from depositors directly in a positive-NPV project.

Banks possess a diversion technology which puts an upper bound on banks' leverage (see Calomiris and Kahn 1991, Hart and Moore 1995, Shleifer and Wolfenzon 2002, Gennaioli and Rossi 2013, and Biswas and Koufopoulos 2022). This friction restricts competition which allows banks to make some profits which cover the cost of effort incurred at the initial stage. Hence, there will be some positive effort exertion in equilibrium. By anticipating the equilibrium in the deposit market, at the ex-ante stage, agents choose the probability with which to exert costly effort to become bankers. This probability is chosen such that agents are indifferent between becoming bankers and depositing in a bank. The probability of effort exertion affects bank entry, which in turn, affects the demand and supply of deposits; that is, the market clearing condition is endogenously determined.

Consider the case that the cost of effort is positive but small enough such that effort exertion is efficient. In equilibrium, the diversion constraint must bind. If it is slack, banks can increase their profits by deviating to higher leverage, while keeping the deposit rate unchanged. The ex-ante effort incentive constraint must also bind. If the effort incentive constraint is slack, agents' expected profits from exerting effort is strictly higher than those from not exerting effort. Hence, at $t = 0$ they exert effort with a higher probability until the effort incentive constraint binds. Thus, the equilibrium is determined at the intersection of the diversion constraint and the agents' effort incentive constraint, and the deposit market clears.

If the cost of effort is very high, agents do not exert effort since the incentive constraint is violated and the banking equilibrium breaks down. For a smaller (but positive) cost of effort, banks offer a deposit rate which is lower than the competitive rate under full information which allows them to make profits such that they recover the cost of exerting effort. If the cost of effort is zero, the deposit rate coincides with the competitive rate under full information, implying that depositors extract the full surplus.

The distinguishing feature of our model is as follows: despite price competition in the deposit market, banks retain some pricing power allowing them to offer a deposit

rate which is lower than the rate they would have offered under full pledgeability. This enables banks to just recover the cost of effort incurred at the ex-ante stage. In contrast to existing models where competition prevents the recovery of sunk costs, our model incorporates a binding diversion constraint that limits competition, ensuring the recovery of the sunk cost. While this constraint restricts profits by preventing banks from leveraging up infinitely, it also serves as the friction that facilitates bank entry when entry is costly. Without the diversion constraint, agents would not exert effort, even if the cost is infinitesimally small (but strictly positive), as they would not be able to recover the cost of effort. This lack of effort leads to a collapse of the banking equilibrium. In this case, a lower cost of entry does not affect the equilibrium unless the cost is reduced to zero. Thus, the conventional wisdom in industrial organization, that lower entry barriers promotes entry, holds in our model only when pledgeability is imperfect.

Empirical predictions and policy implications.

The interaction of the diversion constraint with the ex-ante effort moral hazard constraint produces a number of new implications:

The effect of an increase in pledgeable income on bank leverage is not necessarily positive. If pledgeable income increases due to institutional factors which allows banks to commit to diverting a smaller fraction of the output, then it leads to higher leverage. However, if pledgeable income increases due to improved investment opportunities, then it leads to a lower leverage. The latter effect arises as improved investment opportunities positively affects banks' entry in the initial stage. As the number of banks in the sector increase, each bank takes on lower leverage. We present several other new predictions in Section 3.1.

Besides generating new empirical predictions, our model also offers interesting insights regarding the measurement of the degree of bank competition. Numerous empirical studies use concentration-based proxies (such as the Herfindahl-Hirschman index) to measure bank competition. However, consistent with evidence presented in Berger and Hannan (1989) and Hannan (1992), our model implies that the concentration-based measures of bank competition may be negatively correlated with the margin-based measures such as the Lerner index (see also, Claessens and Laeven 2004 and Shaffer 1993).

Hence, the use of concentration-based proxies in empirical studies can lead to misleading interpretations unless the degree of diversion is appropriately accounted for.

In terms of policy implications, we show that when the cost of effort falls within an intermediate range, bank entry is excessive in the unregulated equilibrium. This contrasts the standard result in industrial organization that entry is too little in the presence of entry barriers. An utilitarian regulator can step in to eliminate this excessive entry inefficiency. One effective measure is the introduction of an upper bound on the number of banking licenses issued.

Related literature.

Our paper is related to models of bank competition and financial stability (see, e.g., Hellmann et al. 2000, Repullo 2004, Martinez-Miera and Repullo (2010), Allen et al. (2011), Carletti and Leonello (2019), and Carletti et al. (2023)). However, in all of these models, the degree of competition is exogenously determined. In contrast, we derive the degree of competition in the deposit market as an endogenous outcome based on the model's deep parameters. Specifically, the binding diversion constraint determines the level of competition, which intensifies as pledgeability increases.

In several models, asymmetric information on the asset side endogenously restricts competition in the banking sector which enables banks to earn informational rents (see, e.g., Broecker 1990, Rajan 1992, Shaffer 1998, Dell'Ariccia et al. 1999, Boot and Thakor 2000, Dell'Ariccia 2001, Marquez 2002, Hauswald and Marquez 2003, Dell'Ariccia and Marquez 2004, and Hauswald and Marquez 2006). In contrast, we model the funding side of banks.

Several models of imperfect competition in banking study how monetary policy shocks affect deposit franchise values and their transmission to the real economy (e.g., Drechsler et al. 2017, Drechsler et al. 2021, Wang et al. 2022, Drechsler et al. 2023). Our model introduces a channel through which shocks to banks' investment opportunities is transmitted to the liability side via the banks' entry condition. Specifically, a positive shock to investment opportunities results in higher deposit rates, but as bank entry increases and competition intensifies, leverage falls. By endogenizing the source of banks' market power our approach offers richer predictions, particularly when accounting for banks'

entry decision. For example, while the partial equilibrium effect of an improvement in investment opportunities is higher leverage, there is also an entry effect which works in the opposite direction, and this latter effect strictly dominates the former. Thus, the partial equilibrium prediction is reversed when the bank's entry decision is taken into account.

Our work is also related to models in industrial organization where price competition does not necessarily lead to zero profits which is the standard Bertrand result. Kreps and Scheinkman (1983) analyze a two-stage game where competing firms first commit to production quantities, creating capacity constraints that soften price competition in the subsequent stage. Unlike in Kreps and Scheinkman (1983), our model introduces capacity constraints arising from an informational friction (the diversion constraint), independent of the game's design. Also, Kreps and Scheinkman (1983) consider a fixed number of firms, whereas we allow for entry. Additionally, Rotemberg and Saloner (1986) show that dynamic strategic considerations in repeated interactions can lead to oligopolistic outcomes despite price competition. However, our model focuses on single-period competition among banks.

2 Model

We develop a model in which a bank raises funds in the deposit market and invests them directly. Following the interpretation in Allen et al. (2015), direct investment by banks may be thought of as fee-generating investment banking activities. Another equivalent interpretation is that banks behave monopolistically in the loan market and extract the full surplus in this market.

2.1 Set-up

We consider an economy in which there is a measure 1 of agents, each with an initial endowment of 1 unit. To begin with, agents are identical. Heterogeneity across agents arises at the next stage in which agents learn whether they have acquired the skill to manage banks or not. Those who acquire the skill to manage banks become bankers and

they manage their own funds and any funds that they raise externally. Agents who do not become bankers either invest in the storage technology, or become depositors, in which case they delegate the management of their funds to bank.

There are three dates, $t = (0, 1, 2)$. At $t = 0$, agents can either exert effort or not, and this choice of effort affects the probability with which they become bankers. Effort exertion costs k . If an agent does not exert effort, the probability with which they become a banker is 0. Effort exertion increases the agent's probability of becoming a banker. We assume that if agents exert effort, they will become bankers with probability 1. A fraction $\lambda \in [0, 1]$ of the agents exert effort, where λ is determined in equilibrium through the optimizing behaviour of agents. Effort exertion only affects the probability of becoming a banker, but it does not affect the quality of the banker, i.e., banks are identical.¹

At $t = 1$, banks raise funds in the deposit market, and directly invest in a project. At $t = 2$, the returns are realized. Per unit of investment, the project yields X if it succeeds, and 0 if it fails. The project exhibits constant returns to scale and is infinitely scalable. A bank succeeds with probability p and fails with the complementary probability, $1 - p$. An agent who did not acquire the skill to manage banks may also invest in this project, but their success probability is lower, $p_l < p$. For this reason, it is optimal for these agents to delegate the management of their funds to banks.

As in Biswas and Koufopoulos (2022), a banker can divert a fraction, $(1 - \phi)$, of the realized output at $t = 2$, $\phi \in (0, 1)$. Diversion of funds is not verifiable, and hence, the banker cannot credibly commit ex-ante to not diverting these funds even if she would like to be able to do so. Thus, our model features two informational frictions: there is an ex-ante effort moral hazard at $t = 0$, and a diversion (or limited pledgeability) constraint at $t = 1$. We make the following parametric assumptions:

$$A1: pX > p_l X = 1$$

Assumption A1 implies that a bank's project is profitable and it has a positive net

¹We assume that it is perfectly observable whether an agent has acquired the skill to manage a bank or not. The interpretation is that all agents take a cost-less exam which produces a precise signal about their ability to manage banks. Even if we make the signal imprecise such that some agents who have exerted effort fail the test, all of our qualitative results are unchanged. A less precise signal would be equivalent to a higher cost of effort, k . However, if an agent does not exert effort he will never pass the test, implying that all banks are identical and there is no adverse selection problem.

present value (NPV). The NPV of projects managed by those who did not acquire the skill to manage funds is normalized to 0. That is, privately investing outside the banking sector is equivalent to having access to a costless storage technology.

$$\text{A2: } \phi < \frac{1}{pX}$$

Assumption A2 implies that diversion is sufficiently large such that there exist strictly positive values of k for which there is effort exertion.

2.2 The Game

We consider the following game:

Stage 1: An agent decides whether or not to exert effort, $e \in \{0,1\}$. $e = 1$ denotes effort exertion and imposes a cost, k , on the agent, while $e = 0$ denotes shirking and the agent saves the cost. If they exert effort, their probability of becoming a banker increases from 0 to 1. In the next stage, agents who become bankers seek external financing in the deposit market.

Stage 2a: Agents who did not become bankers either deposit their funds in banks or invest in the storage technology. A banker offers a contract, (d, R) , to depositors where d is the bank's publicly observable level of leverage, and R is the promised repayment rate.

Stage 2b: After observing the bank's proposal, depositors decide whether to accept the offer or reject it and invest in the storage technology.

Stage 3: Bankers invest their own endowment and any funds externally raised directly in the project.

We look for the sub-game perfect Nash equilibrium of this game. We solve for the equilibrium backwards. First, we derive the structure of the deposit market, taking as given the fraction of agents who become bankers. Then, we derive agents' level of effort exertion in Stage 1 of the game by fully anticipating the equilibrium in the deposit market. Since agents are identical in Stage 1, we consider a mixed strategy equilibrium in which a fraction of agents choose to exert effort by trading off the benefit of becoming a banker against the cost of effort, k .

2.3 Equilibrium

First, we consider the optimization problem of a banker at $t = 1$ (Stage 2a). The banker invests her own funds and the funds raised externally from depositors. Suppose that the amount of funds raised externally is d , implying that a bank's total investment is $(1 + d)$ units. With probability p , these funds produce X (per unit), and they produce 0 with probability $1 - p$. In the event of success (i.e., with probability p), the banker repays dR to depositors. In maximizing her profits, the banker faces four constraints at $t = 1$:

1. the diversion constraint: given diversion, the total expected verifiable cash flow is $\phi(1 + d)pX$. This total expected verifiable cash flow must be weakly greater than the sum of the total expected promised repayments, which comprises the promised return, dpR , to the depositors, and the expected cash flow contractually retained by the banker himself, say, pR_s . Note that the banker's total return is the sum of the contractually agreed amount and the amount that he diverts from the output.
2. the depositors' participation constraint: the depositors must be weakly better off from depositing their funds with the bank, in which case they obtain, pR , than investing in the storage technology. In the latter case, depositors will get $p_l X = 1$ per unit, in expectation.
3. the banker's limited liability constraint: the promised repayment cannot exceed X .
4. the market clearing condition: the demand for deposits by banks, $d\lambda$ (there are banks of a measure λ , each of which demands d depositors), cannot exceed the supply of deposits, $1 - \lambda$ (these are the agents who do not exert effort to acquire the skill to manage banks). Since λ is chosen by agents at $t = 0$, this constraint is endogenous from the perspective of the overall game but taken as given at $t = 1$.

The only endogenous variable to be chosen in Stage 2a are d and R . The banker's problem in this stage of the game is formally stated below:

$$\begin{aligned}
& \text{Max}_{d,R} && p(1+d)X - pdR - 1 \\
& \text{subject to} && \phi p(1+d)X \geq pdR + pR_s \\
& && pR \geq p_l X = 1 \\
& && R \leq X \\
& && d\lambda \leq 1 - \lambda
\end{aligned} \tag{1}$$

For any $R < X$, since the bank retains a fraction of the surplus generated from investing the externally raised funds, bank profits are strictly increasing in the number of external investors, d . Given the observation that bank profits are increasing in d , we solve for the maximum d which is consistent with the diversion constraint being satisfied. We re-write the diversion constraint as follows:

$$(pR - \phi pX)d \leq \phi pX - pR_s \tag{2}$$

The LHS is increasing in d . Hence, to relax this constraint to the maximum extent possible (while not affecting any of the other constraints) we set $R_s = 0$. Thus, external investors have priority over the verifiable fraction of the cash flow, implying that they receive the most senior claim which we refer to as deposit. The banker obtains the junior claim and becomes the residual claimant (the equity-holder). This arrangement is privately optimal for the banker since it allows the bank to attract the maximum amount of funds from outside investors.

Lemma 1 (Optimal contracts) *There are d outside investors in the bank and the optimal contract for the outside investors is debt/deposit, and the banker holds the residual claim which is equity.*

Proof. The proof follows from the text. ■

Setting $R_s = 0$, the diversion constraint becomes:

$$d \leq \frac{\phi X}{R - \phi X} \quad (3)$$

The diversion constraint puts an upper bound on the amount of funds that the depositors are willing to provide to the bank. To derive the slope of the diversion constraint in the (d, R) space, we totally differentiate Equation (3) with respect to d and R :

$$\left. \frac{dd}{dR} \right|_{DC} = -\frac{d}{R - \phi X} < 0 \quad (4)$$

The slope of the diversion constraint is negative (and it is concave). This reflects that as each depositor is promised a higher repayment, the bank can accept fewer deposits since the diversion constraint tightens. A feasible equilibrium must lie below the diversion constraint.

Lemma 2 *For any λ , the equilibrium in the deposit market is such that diversion constraint must bind in equilibrium for any $R < X$.*

Proof. At $t = 1$, λ is taken as given since it is determined in the previous stage of the game. Thus, at this stage when banks compete with each other over deposits, the relevant constraint is the diversion constraint. Given this, regardless of the choice of λ in the previous stage, the diversion constraint must bind. If it is slack, a bank can profitably deviate by accepting more deposits by offering an ϵ higher deposit rate (where ϵ is positive but arbitrarily small), which attracts depositors away from other banks till its diversion constraint binds. This deviation is feasible as long as $R < X$, since when $R = X$, the bank's liability constraint binds, and it cannot offer an ϵ higher deposit rate. ■

The depositors' participation constraint, $R = \frac{1}{p} = R^P$, and the banker's limited liability constraint, $R = X$ are vertical lines in the (d, R) space. Given that the bank-managed projects are profitable (Assumption A1), the depositors' participation constraint lies to the left of the limited liability constraint. A feasible equilibrium must lie to the right of the depositors' participation constraint and to the left of the limited liability constraint.

The market clearing condition is a horizontal line in the (d, R) space. For very small effort, i.e. when λ is small, the market clearing condition lies above the intersection of the diversion constraint and the depositors' participation constraint. Higher effort exertion pushes the market clearing condition down in the (d, R) space. A feasible equilibrium must lie below the market clearing condition. In stage 2 of the game, this constraint is taken as given, but in Stage 1, agents choose λ by fully anticipating the equilibrium d and R which are determined in the next stage. That is, from the perspective of $t = 0$, the market clearing condition is endogenously determined.

The effort incentive constraint should be considered in the first stage of the game at which point agents exert effort to determine the probability with which they become bankers. Since exerting effort is assumed to be costly, agents exert effort only if they are (weakly) better off from exerting effort than not. By exerting costly effort, agents ensure that they become bankers with probability 1 (LHS, below). If they do not exert effort, they deposit their endowment in a bank (RHS, below).² The incentive constraint is as follows:

$$(1 + d)pX - dpR - 1 - k \geq pR - 1 \quad (5)$$

Re-writing the incentive constraint, an agent exerts effort if the cost of effort is sufficiently small:

$$k \leq (1 + d)p(X - R) \quad (6)$$

The effort exertion decision of agents is undertaken at $t = 0$ by fully anticipating the outcomes, d and R , in Stage 2 of the game. We consider mixed strategy equilibria in which a fraction of agents exert effort and the rest do not. Starting with a set of identical agents, some agents exert effort and others do not only if their payoffs are equal in equilibrium, implying that they are indifferent between exerting effort or not.

To derive the slope of the incentive constraint in the (d, R) space, we totally differenti-

²The agent has the choice to invest in the storage technology as well. However, the deposit rate R is (weakly) higher than the return on the storage technology, R_P , and hence, without loss of generality, we can only consider deposit as the alternative.

ate Equation (6) with respect to d and R :

$$\left. \frac{dd}{dR} \right|_{IC} = \frac{1+d}{X-R} > 0 \quad (7)$$

The slope of the incentive constraint is positive (and it is convex). This reflects that as agents anticipate that depositors will be promised a higher repayment in the future, their incentive to exert effort at $t = 0$ to become bankers becomes weaker for two reasons: when R increases, first, banks keep less of the surplus, and second, depositors obtain more of it. As R approaches X , the slope becomes very steep, implying that the incentive constraint gets arbitrarily close to the $R = X$ vertical line, but does not cross it. A feasible equilibrium must lie on or above the incentive constraint.

Increasing the cost of exerting effort, k , makes the moral hazard constraint tighter, but it does not affect any of the other constraints. The intersection of the effort incentive constraint and the diversion constraint depends on the cost of effort. The lower the cost, the intersection of the two constraints is further to the right in the (d, R) space. There are four cases to consider:

1. $k > \frac{pX-1}{1-\phi pX} \equiv \bar{k}$
2. $k = \bar{k}$
3. $\bar{k} > k > 0$
4. $k = 0$

2.3.1 Case 1: $k > \bar{k}$

We illustrate the depositors' participation constraint, the diversion constraint, and the limited liability constraint, and the incentive constraint for $k > \bar{k}$ in Figure 1. In this case, the incentive constraint intersects the diversion constraint to the left of the depositors' participation constraint. Agents exert effort for a (d, R) pair which lies above the incentive constraint. It can be seen that the set of parameters where agents exert effort, and at the same time, none of the exogenous constraints is violated, is empty. Thus, in equilibrium,

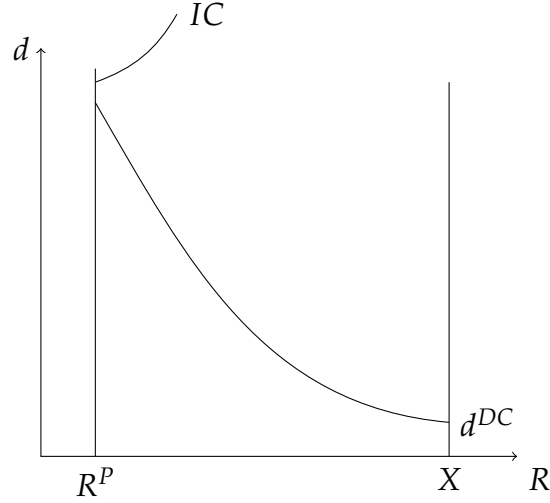


Figure 1: No banks

there cannot be effort exertion, which implies that there are no banks. We summarize this results in the proposition below:

Proposition 1 (No banks) *Suppose that $k > \bar{k}$. No agent exerts effort implying that there are no banks, and agents invest on their own.*

Proof. The proof follows from the text. ■

Provided that depositors get at least R^P , the payoff for any agent who exerts effort is strictly less than the cost of effort (i.e., the net surplus from effort exertion is strictly negative). Thus, no agent will exert effort because investing in the storage technology makes them strictly better off.

2.3.2 Case 2: $k = \bar{k}$

In Figure 2, we consider the $k = \bar{k}$ case. Now, the incentive constraint goes through the intersection of the depositors' participation constraint and the diversion constraint (the

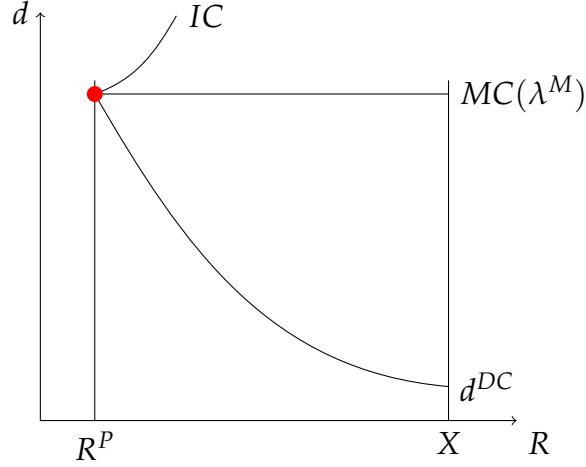


Figure 2: Monopoly pricing

red dot), which is characterized as follows:

$$d = \frac{\phi p X}{1 - \phi p X} \equiv d^M \quad (8)$$

$$R = \frac{1}{p} = R^P \quad (9)$$

Suppose that the market clears. To derive the level of effort exertion at the red dot, we substitute $d = d^M$ and $R = R^P$ in the market clearing condition, which gives:

$$\lambda = 1 - \phi p X \equiv \lambda^M \quad (10)$$

Even for a lower level of effort exertion, $\lambda < \lambda^M$, the only feasible equilibrium is at the red dot and there is no deviation from this equilibrium in which agents make strictly higher profits. Thus, any λ less than λ^M is also an equilibrium, and the level of effort exertion cannot be pinned down. This implies that, for $k = \bar{k}$, the market may or may not clear (in which case there is an excess supply of deposits). We summarize these results in the proposition below:

Proposition 2 (Monopoly pricing) *Suppose that $k = \bar{k}$. The level of effort exertion lies in the range, $\lambda \in [0, \lambda^M]$. Although banks compete in prices, they offer the monopoly deposit rate. The equilibrium features $d = d^M$ and $R = R^P$.*

Proof. The proof is in the Appendix. ■

In this equilibrium, the depositors are on their participation constraint. Hence, we refer to this as the monopoly pricing equilibrium even though there is a continuum of banks in equilibrium. Although banks in this case charge monopoly prices from the $t = 1$ perspective, taking into account the cost of exerting effort, they make zero profits from the $t = 0$ perspective. Indeed, under the monopoly outcome with λ^M banks, the total cost incurred by banks is $\lambda^M \bar{k} = pX - 1$, which equals the NPV of banks' investment. That is, the banking equilibrium produces a zero net surplus in this case. This is why the equilibrium cannot be uniquely determined since the payoff for all agents, regardless of whether they exert effort or not, is the exactly the same as if they invest on their own in the storage technology.

2.3.3 Case 3: $\bar{k} > k > 0$

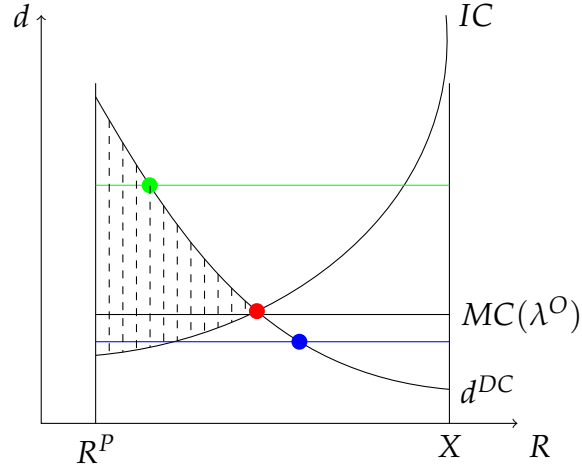


Figure 3: Oligopoly pricing

In Figure 3, we consider the $\bar{k} > k > 0$ case. In this case, the incentive constraint intersects the diversion constraint in the space between the depositors' participation constraint and the banker's limited liability constraint and this intersection is represented by the red dot. The feasible equilibria with effort exertion are in the shaded area. We argue below that in equilibrium λ would be chosen such that the market clearing condition goes

through the red dot.

We know from Lemma 2 that from the $t = 1$ perspective, regardless of the choice of λ in the previous stage, the equilibrium in the deposit market is such that the diversion constraint binds. Suppose that λ is chosen such that the market clearing condition lies above the red dot (the green line). In this sub-game, the equilibrium is at the intersection of the market clearing condition and the diversion constraint (the green dot). However, from the $t = 0$ perspective, this cannot be an equilibrium since the incentive constraint is slack at the green dot (the expected payoff from exerting effort is strictly greater than not). This implies that agents will exert effort with a higher probability, and λ increases until the incentive constraint binds and the market clearing line goes through the red dot. Now, consider the sub-game in which λ is chosen such that the market clearing condition lies below the red dot (the blue line). In this sub-game, the equilibrium in the deposit market is at the intersection of the market clearing condition and the diversion constraint (the blue dot). However, since the blue dot is below the incentive constraint, banks make expected losses from the $t = 0$ perspective. Anticipating this outcome at $t = 0$, agents will reduce their probability of exerting effort at the initial stage, which pushes the market clearing line up till the incentive constraint is satisfied, i.e., it reaches the red dot.

Thus, the only value of λ which can be consistent with equilibrium is such that the market clearing line goes through the red dot. Now we argue that this is indeed an equilibrium. First, note that in this subgame, the equilibrium is at the red dot since the diversion constraint must bind. From the $t = 0$ perspective, agents obtain the same expected payoff whether they become a banker or a depositor at this red dot. By deviating to a different λ , an agent cannot improve his payoff over what he obtains in the equilibrium at the red dot. To see this, consider a deviation where in the deposit market the agent offers a lower deposit rate. Depositors will not accept a lower deposit rate when they can get the rate at the red dot and hence this is not a profitable deviation. Nor can a bank profitably deviate by offering a higher deposit rate since, because of the diversion constraint, in this case it will be able to attract fewer depositors, leading to an unambiguous fall in its profits. Thus, there is no profitable deviation, and hence, the equilibrium at the red dot exists. Further, since no other λ can be an equilibrium, this equilibrium is unique.

We summarize these results in the proposition below:

Proposition 3 (Oligopoly pricing) *Suppose that $\bar{k} > k > 0$. Agents exert effort with probability λ^O . The unique equilibrium is characterized by (λ^O, d^O, R^O) where:*

$$\lambda^O = \frac{(pX + k) - \sqrt{(pX + k)^2 - 4kpX(1 - \phi)}}{2k} \quad (11)$$

$$d^O = \frac{1 - \lambda^O}{\lambda^O} \quad (12)$$

$$R^O = \frac{\phi X}{1 - \lambda^O} \quad (13)$$

Proof. The proof is in the Appendix. ■

We refer to this as the oligopoly pricing equilibrium since while depositors are better off compared to their outside option, banks offer a deposit rate which is lower than the competitive rate under full information, i.e., $R^P < R^O < X$.

2.3.4 Case 4: $k = 0$

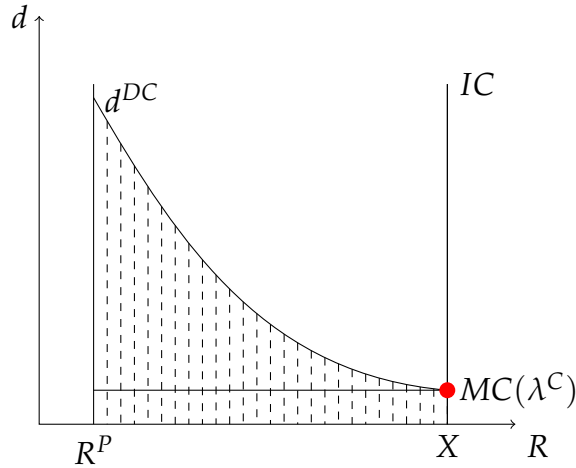


Figure 4: Competitive pricing

In Figure 4, we consider the $k = 0$ case. For $k = 0$, the incentive constraint (Equation (5)) reduces to $R = X$, i.e., it coincides with the limited liability constraint. The feasible

equilibria with effort exertion are in the shaded area and it is the entire region between the depositors' participation constraint and the limited liability constraint, below the diversion constraint.

First, we show that an equilibrium with $R < X$ is not stable. If $R < X$ holds, then the incentive constraint at $t = 0$ is slack. In this case, agents can profitably deviate by exerting higher effort until the market clearing condition goes through the intersection of the diversion constraint and the limited liability constraint (the red dot). That is, competition drives the deposit rate up to the maximum feasible level, and we obtain the Bertrand outcome, $R = X$. At the red dot, all constraints are satisfied and there is no strictly profitable deviation from this point. Hence, this is an equilibrium. Substituting $R = X$ into the diversion constraint, we get:

$$d = \frac{\phi}{1 - \phi} \equiv d^C \quad (14)$$

Substituting $d = d^C$ into the market clearing condition, we derive the level of effort exertion at the red dot equilibrium:

$$\lambda = 1 - \phi \equiv \lambda^C \quad (15)$$

Beyond this point, the agent is indifferent between exerting more effort or not. Exerting more effort will push down the market clearing condition below the red dot, in which case d will be lower than d^C . Anywhere on the limited liability constraint below the red dot is also an equilibrium since there are no strictly profitable deviations from these points - a bank cannot offer depositors a higher interest rate due to the binding limited liability constraint. Thus, for $k = 0$, while the deposit rate is uniquely determined, bank leverage cannot be pinned down. The probability with which an agent exerts effort is $\lambda \in [\lambda^C, 1]$ and the corresponding bank leverage is such that the market clears. We summarize these results in the proposition below:

Proposition 4 (Competitive pricing) *Suppose that $k = 0$. The level of effort exertion is $\lambda \geq \lambda^C$. The market for deposits is perfectly competitive and the equilibrium features $(d \leq d^C, R =$*

X). The (λ, d) pair is such that the market clears.

Proof. The proof follows from the text. ■

We refer to this as the competitive pricing equilibrium since depositors extract the full surplus generated from the investment of their endowment. For a similar reason as the $k = \bar{k}$ case, there is multiplicity of equilibria when $k = 0$. Specifically, the expected payoff for all agents, regardless of whether they become bankers, depositors, or invest on their own (after costlessly acquiring the skill to manage funds), is the exactly the same, $pX - 1$.

2.3.5 Discussion

All else given, expected bank profits are increasing in bank leverage, but the diversion constraint puts an upper bound on the leverage. If the cost of effort is very high, $k > \bar{k}$, agents do not exert effort, since, even for the minimum feasible deposit rate, bank leverage would be too low for the agents to fully recover their cost of effort. For a lower cost of effort, $k \in [0, \bar{k}]$, the equilibrium features effort exertion at $t = 0$. Banks retain pricing power in the deposit market which compensates for the cost of effort incurred in the previous stage (zero profits when $k = 0$). Counter to intuition, banks recover the cost of effort even though this cost is sunk by the time they compete for deposits. What allows banks to recover the sunk cost of effort is the binding diversion constraint. Due to the diversion constraint, agents anticipate pricing power at $t = 1$, and exert effort to the extent that they expect to be compensated for it. Effectively, the binding diversion constraint which limits bank leverage and profits also chokes competition, which, in turn, relaxes the effort incentive constraint in the earlier stage. In the absence of the diversion constraint, banks would never exert costly effort as competition would drive the deposit rate to X and the cost of effort would not be recovered. The perfect competition case in which banks do not possess pricing power in the deposit market arises only for the knife-edge case of zero cost of effort, $k = 0$.

3 Implications

In this section, we present the empirical and policy implications of the model which arise from the interaction of the diversion constraint and the ex-ante effort moral hazard constraint.

3.1 Comparative statics

In this section, we present the new empirical predictions arising out of the model. Consider the case that the cost of effort is in the intermediate range, $0 < k < \bar{k}$, implying that banks retain a part of the surplus, $R < X$. To derive the first prediction, we state the following Lemma:

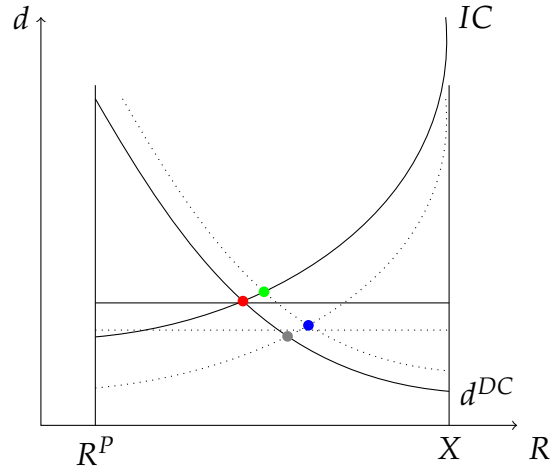


Figure 5: Comparative Statics

Lemma 3 *Bank leverage, d , falls in investment opportunities, X .*

Proof. The proof is in the Appendix. ■

When the bank's investment opportunities improve, which is captured by a higher X , there are two forces at play:

First, there is a direct effect which is that it pushes the diversion constraint out. Taking the other constraints as given, the new equilibrium would move from the red dot to the

intersection of the new dotted diversion constraint with the pre-existing market clearing line in Figure 5. This new equilibrium features a higher deposit rate, but leverage is unaffected. We would expect this to be the relevant comparative static if the increase in X is unanticipated. However, for an anticipated positive shock to X , there is also an indirect general equilibrium effect which is that it relaxes the effort incentive constraint and pushes it down. Both forces have a positive effect on the deposit rate, which increases as a result of a higher X . Interestingly, the two forces have conflicting effects on leverage – the direct effect boosts leverage, taking as given the number of banks operating, but the indirect effect boosts bank entry, and it pushes up λ , which leads to a lower bank leverage. We show in the proof of Lemma 3 that the indirect effect dominates, and the new equilibrium features a lower d when X increases – in Figure 5, the equilibrium moves from the red to the blue dot. That is, even though banks' investment opportunities improve allowing banks to attract more deposits (all else given), anticipating this, more agents enter the banking sector, which intensifies competition and leads to a lower equilibrium leverage.

Prediction 1: *An increase in banks' investment opportunities, X , leads to more banks, a higher deposit rate, R , and a lower bank leverage, d .*

When pledgeable income increases because diversion, $1 - \phi$, falls, the diversion constraint is pushed out, while the other constraints are unaffected. The equilibrium moves up along the incentive constraint implying that fewer agents exert effort at the ex-ante stage, and it features a higher R and a higher d – in Figure 5, the equilibrium moves from the red to the green dot. Intuitively, a higher ϕ implies that banks can pledge more of their output to depositors which leads to higher deposit rates due to competition. Then, given the starting leverage and probability of exerting effort, banks become less profitable which, in turn, results in the violation of the incentive constraint. In response, agents exert effort with a lower probability at the initial stage till the market clearing line goes through the new intersection of the diversion constraint and the incentive constraint (the green dot).

If we assume that the strength of institutions is increasing in GDP/capita, then richer

countries are characterized by a lower threat of diversion. Given this, the prediction is that bank pricing power is falling in GDP/capita, but the banking sector is less concentrated (a higher number of banks). We explore this divergence between different proxies of bank competition further in the next section.

Thus, when pledgeable income increases, the impact on the deposit rate is unambiguously higher regardless of if X or ϕ contributes to the increase in pledgeable income – higher pledgeability leads to lower market power in the deposit market. However, the impact of a higher pledgeable income on bank leverage may be positive or negative depending upon whether ϕ or X contributes to the increase in pledgeable income. If pledgeable income increases due to institutional factors which allows banks to commit to diverting a smaller fraction of the output, then it leads to higher leverage. However, if pledgeable income increases due to improved investment opportunities, then it leads to a lower leverage.

Prediction 2: *An increase in banks' pledgeability, ϕ , leads to fewer banks, a higher deposit rate, R , and a higher bank leverage, d .*

When the cost of bank entry, k , falls, it relaxes the effort incentive constraint and pushes it down, while the other constraints are unaffected. The expected profits from exerting effort becomes higher than the expected profits from not exerting effort, resulting in agents increasing the probability with which they exert effort, and hence, fewer banks. They do so till the market clearing condition reaches the intersection of the new incentive constraint with the diversion constraint. The equilibrium moves down along the diversion constraint and features a higher R and a lower d – in Figure 5, the equilibrium moves from the red to the gray dot.

This prediction aligns with the intuition that lowering barriers to entry encourages entry. However, this result only holds in our model when the diversion constraint is present. To illustrate this, suppose that $\phi = 1$ (violating Assumption A2), meaning that banks cannot divert output. In this case, banks can lever up infinitely, and competition for deposits will drive the deposit rate to $R = X$. Consequently, at $t = 0$, agents will not exert effort

for any strictly positive value of k (even if k is infinitesimally small) since the sunk cost of effort cannot be recovered. Therefore, in the absence of the diversion constraint, reducing k does not necessarily lead to more entry unless k is reduced to zero.

Prediction 3: *A fall in the cost of bank entry, k , leads to more banks, a higher deposit rate, R , and a lower bank leverage, d .*

3.2 Measuring market power

High concentration in the banking sector can coexist with low pricing power.

A higher ϕ relaxes the diversion constraint, pushing it upwards, while leaving the other constraints unaffected. As shown in Figure 5, the equilibrium shifts from the red to the green dot. This new equilibrium is characterized by fewer banks, and yet, a smaller interest rate spread. Conversely, starting from the oligopoly case, if ϕ falls sufficiently, we reach the monopoly outcome, where the diversion and incentive constraints intersect on the vertical line at R_p . This equilibrium features more banks relative to the starting point of oligopoly, and yet, banks capture the full surplus from investment. Thus, a negative correlation between the interest rate spread and the number of banks can emerge.

This discussion suggests that the concentration-based measures of bank competition (such as the number of banks or the Herfindahl-Hirschman Index, HHI, or the n -firm Concentration ratio, CR_n) may not always align with the margin-based measures (like the Lerner Index or the Panzar-Rosse H-Statistic). Consistent with this prediction, Berger and Hannan (1989) present evidence that deposit rates are increasing in banking sector concentration (see also Hannan 1992). The existing explanation for this puzzling observation is that higher concentration leads to lower costs induced by improved efficiency, and a portion of this cost saving is passed on to depositors. In our model, concentration is endogenously determined and it does not affect efficiency. Claessens and Laeven (2004) and Shaffer (1993) are further examples of empirical papers which document a negative correlation between bank concentration and pricing power. Many empirical banking studies

use the concentration-based proxies as indicators of market power. However, our model indicates that to correctly interpret concentration-based proxies of competition, the level of diversion should be appropriately accounted for.

3.3 Optimal policy

A utilitarian regulator can eliminate a problem of excessive bank entry by introducing an upper bound on the number of bank licences to be allocated.

The net social surplus in the banking equilibrium is $\Pi = (pX - 1) - \lambda k$, which is the NPV of the project net of the total cost incurred by banks. $\Pi = 0$ in the monopoly case with $k = \bar{k}$ and $\Pi = pX - 1$ in the perfect competition case with $k = 0$. In the oligopoly case which arises for intermediate values of k , the net social surplus is strictly positive but monotonically falling in k .³ That is, as k falls, the deposit market becomes increasingly more competitive and produces a higher net social surplus. However, for a given k , efficiency obtains when banks are monopolistic.

Note that when the effort level is efficient, there is a minimum number of bankers such that the market for deposits clears – this ensures that the entire initial endowment of the economy is managed by banks, and at the same time, this arrangement minimizes the exertion of costly effort. This outcome arises for $k = \bar{k}$ which is the monopoly case when agents exert effort with probability $\lambda \in [0, \lambda^M]$. When the effort exertion probability is at its upper bound, $\lambda = \lambda^M$, there are $1 - \phi pX$ banks. For $0 < k < \bar{k}$, there is inefficient duplication of effort which leads to excessive bank entry, $\lambda^O > 1 - \phi pX$. A regulator whose objective is to maximize the net social surplus can eliminate the inefficiency by announcing a two step mechanism to allocate banking licences. In the first step, the regulator will choose a maximum of $1 - \phi pX$ agents to apply for a banking licence on a first-come-first-served basis. In the second step, the chosen agents will need to pass an exam to obtain the license. Since $k < \bar{k}$, the incentive constraint is slack and all $1 - \phi pX$ chosen agents exert effort and acquire the banking skill which enables them to

³To see this, note that $\frac{\partial \Pi}{\partial k} = -\frac{1}{2} + \frac{(pX+k)-2pX(1-\phi)}{2\sqrt{(pX+k)^2-4kpX(1-\phi)}}$ and the derivative is negative for any $\phi > 0$.

pass the exam. Thus, the problem of excessive effort provision is eliminated since there are just enough banks in equilibrium such that the market clears. This regulated equilibrium resembles the unregulated monopoly equilibrium which arises when $k = \bar{k}$ in terms of the outcome variables, d and R . However, in contrast to the laissez faire monopoly, the agents who become bankers in the regulated monopoly (i.e., those agents in the front of the queue who are picked by the regulator to enter the licence exam) extract rents from depositors.

As opposed to the charter value hypothesis (see, e.g., Hellmann et al. 2000) in which higher market power promotes stability by making banks more prudent in anticipation of higher charter values, in our theory, restricting competition in the banking sector improves efficiency by mitigating an excessive effort provision problem.

4 Conclusion

We develop a model of bank competition in the deposit market in which there are two key frictions at play – an ex-post threat of diversion and an ex-ante effort moral hazard friction. The threat of diversion prevents banks from leveraging up infinitely which enables them to recover the sunk cost associated with effort exerted at the initial stage. This allows banks to make positive profits in the deposit market despite price competition. Competition intensifies as diversion falls. The ex-ante effort moral hazard constraint determines the intensity of entry into the banking sector.

The interaction of the diversion constraint and the ex-ante effort moral hazard constraint generates novel implications: While an increase in banks' investment opportunities allows them to increase leverage in a static sense, it also relaxes banks' entry condition. Taking into account the latter effect, the banking sector features more banks and each bank exhibits lower leverage. High concentration in the banking sector can coexist with low pricing power, implying that the use of concentration-based proxies for market power can be misleading unless the level of diversion is appropriately accounted for in empirical studies. In terms of policy implications, there may be excessive entry into the banking sector despite a strictly positive cost of entry. In this case, the regulator intro-

duces an upper bound on the number of bank licences to be allocated which eliminates the inefficiency.

5 Appendix

Proof of Proposition 2.

Proof. We illustrate the various exogenous constraints and the incentive constraint in Figure 2. For $k = \bar{k}$, the incentive constraint goes through the intersection of the depositors' participation constraint and the diversion constraint (the red dot). This is the only (d, R) pair where all the exogenous constraints as well as the incentive constraint are satisfied.

We know from Lemma 2 that, from the $t = 1$ perspective, regardless of the choice of λ in the previous stage, the diversion constraint must bind.

Suppose that the equilibrium is at the red dot and the deposit market clears, i.e, the market clearing condition holds with equality. To derive the equilibrium level of effort exertion, we substitute $d = d^M$ and $R = R^P$ in the market clearing condition, which gives $\lambda = \lambda^M$. This is an equilibrium since there is no profitable deviation from this point given that both the diversion and incentive constraints are binding.

Now consider the case that agents exert a lower level of effort, $\lambda \in (0, \lambda^M)$ such that the market clearing condition is above the red dot. Noting that the diversion constraint must bind, the red dot is still the only feasible equilibrium, albeit in this case, the market does not clear and there is an excess supply of deposits. For $\lambda < \lambda^M$, the incentive constraint is slack but agents cannot profitably deviate by exerting higher effort since the binding diversion constraint prevents increasing the leverage and the binding participation constraint of depositors prevents lowering the deposit rate. Thus, it cannot be ruled out that λ is chosen such that the market clearing condition lies above the red dot.

Finally, suppose that λ is chosen such that the market clearing condition lies below the red dot. Since the diversion constraint must bind, the candidate equilibrium is at the intersection of the diversion constraint and the market clearing condition. However, since the incentive constraint is violated at this point, banks make expected losses in this equilibrium from the $t = 0$ perspective. Anticipating this outcome at $t = 0$, agents will

reduce their probability of exerting effort at the initial stage, which pushes the market clearing line up till it reaches the red dot.

To summarize, the red dot is the unique equilibrium of this game, although the probability of effort exertion cannot be uniquely determined. ■

Proof of Proposition 3

Proof. The text above the proposition establishes that the unique equilibrium is at the red dot. Here, we derive the expressions. The intersection of the diversion constraint and the market clearing condition occurs:

$$d = \frac{(1 - \lambda)}{\lambda} \quad (16)$$

$$R = \frac{\phi X}{1 - \lambda} \quad (17)$$

To pin down λ , we substitute Equations (16) and (17) into the incentive constraint which gives the following quadratic equation:

$$\lambda^2 k - \lambda(pX + k) + (1 - \phi)pX = 0 \quad (18)$$

Using the quadratic formula, we derive the level of effort, λ^O (see Equation (11)). The equilibrium d and R are obtained by substituting λ^O in Equations (16) and (17), respectively.

We show below that:

1. a real solution to the quadratic equation always exists,
2. the bigger root is discarded since it is greater than 1 which λ cannot be,
3. the smaller root always lies below 1, and
4. the smaller root is always positive.

A real solution exists if the discriminant is positive:

$$\begin{aligned}
& (pX + k)^2 - 4kpX(1 - \phi) \geq 0 \\
& \implies (pX + k)^2 \geq 4kpX(1 - \phi) \\
& \implies p^2X^2 + k^2 + 2kpX \geq 4kpX - \phi 4kpX \\
& \implies p^2X^2 + k^2 - 2kpX \geq -\phi 4kpX \\
& \implies (pX - k)^2 \geq -\phi 4kpX
\end{aligned}$$

Since the LHS is positive and the RHS is negative, this condition is satisfied, implying that a real solution always exists.

The bigger root is greater than 1 if:

$$\begin{aligned}
& (pX + k) + \sqrt{(pX + k)^2 - 4kpX(1 - \phi)} \geq 2k \\
& \implies \sqrt{(pX + k)^2 - 4kpX(1 - \phi)} \geq 2k - (pX + k) \\
& \implies (pX + k)^2 - 4kpX(1 - \phi) \geq (k - pX)^2 \\
& \implies k^2 + p^2X^2 + 2kpX - 4kpX + 4\phi kpX \geq k^2 + p^2X^2 - 2kpX \\
& \implies \phi pX \geq 0
\end{aligned}$$

Since the above condition is satisfied, the bigger root is always greater than 1, and hence, it can be discarded.

The smaller root is smaller than 1 if:

$$\begin{aligned}
& (pX + k) - \sqrt{(pX + k)^2 - 4kpX(1 - \phi)} \leq 2k \\
& \implies (pX + k) - 2k \leq \sqrt{(pX + k)^2 - 4kpX(1 - \phi)} \\
& \implies (pX - k) \leq \sqrt{(pX + k)^2 - 4kpX(1 - \phi)} \\
& \implies (pX - k)^2 \leq (pX + k)^2 - 4kpX(1 - \phi) \\
& \implies p^2X^2 + k^2 - 2kpX \leq p^2X^2 + k^2 + 2kpX - 4kpX + 4\phi kpX \\
& \implies \phi pX \geq 0
\end{aligned}$$

Since the above condition is satisfied, the smaller root is always smaller than 1.

The smaller root is greater than 0 if:

$$\begin{aligned}
& (pX + k) - \sqrt{(pX + k)^2 - 4kpX(1 - \phi)} \geq 0 \\
& \implies (pX + k) \geq \sqrt{(pX + k)^2 - 4kpX(1 - \phi)} \\
& \implies (pX + k)^2 \geq (pX + k)^2 - 4kpX(1 - \phi) \\
& \implies (1 - \phi)pX \geq 0
\end{aligned}$$

Since the above condition is satisfied, the smaller root is always positive. ■

Proof of Lemma 3.

Proof. Totally differentiating the diversion constraint with respect to R and X :

$$\left. \frac{dR}{dX} \right|_{DC} = \frac{\phi(1 + d)}{d} \quad (19)$$

Totally differentiating the incentive constraint with respect to R and X :

$$\left. \frac{dR}{dX} \right|_{IC} = 1 \quad (20)$$

For a given increase in X , both the diversion and incentive constraints shift to the right in

the (d, R) space indicating that a higher X leads to an increase in R . To assess the impact of a higher X on leverage and entry, we compare the magnitude of the lateral shift in the two constraints. The magnitude of the lateral shift in the diversion constraint is smaller than that of the incentive constraint if:

$$\begin{aligned} \frac{\phi(1+d)}{d} &< 1 \\ \implies d &> \underbrace{\frac{\phi}{1-\phi}}_{d^C} \end{aligned}$$

The diversion constraint is downward sloping, and for any $k > 0$, it always binds. When $k = 0$, $d = d^C$ (the perfect competition case). Hence, the above condition holds for any $k > 0$ (the oligopoly and monopoly cases). Given this observation that an increase in X leads to a smaller lateral shift in the diversion constraint than in the incentive constraint, when X increases, the diversion and incentive constraints intersect at a lower point. This implies that a higher X leads to more banks and a lower d . ■

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